Plan d'exécution d'une requête

Commande : explain

```
bpinaud=# explain select * from repas natural join dragons where sexe='M';
                              QUERY PLAN
-----------------------------------------------------------------------
 Hash Join  (cost=14.90..30.34 rows=2 width=262)
   Hash Cond: ((repas.dragon)::text = (dragons.dragon)::text)

   ->  Seq Scan on repas  (cost=0.00..14.30 rows=430 width=160)

   ->  Hash  (cost=14.88..14.88 rows=2 width=180)
         ->  Seq Scan on dragons  (cost=0.00..14.88 rows=2 width=180)
               Filter: (sexe = 'M'::bpchar)
```

Plan d'exécution d'une requête

Commande : explain

```
bpinaud=# explain select * from repas natural join dragons where sexe='M';
                              QUERY PLAN
-----------------------------------------------------------------------
 Hash Join  (cost=14.90..30.34 rows=2 width=262)
   Hash Cond: ((repas.dragon)::text = (dragons.dragon)::text)
   ->  Seq Scan on repas  (cost=0.00..14.30 rows=430 width=160)
   ->  Hash  (cost=14.88..14.88 rows=2 width=180)
         ->  Seq Scan on dragons  (cost=0.00..14.88 rows=2 width=180)
               Filter: (sexe = 'M'::bpchar)
```

Estimation du coût d'exécution (en unité de coût, nombre de pages de disque lues)
Premier nombre : estimation du coût avant de retourner le premier n-uplet
Deuxième nombre : coût total pour retourner tous les n-uplets

Plan d'exécution d'une requête

Commande : explain

```
bpinaud=# explain select * from repas natural join dragons where sexe='M';
                              QUERY PLAN
-----------------------------------------------------------------------
 Nested Loop  (cost=4.16..10.64 rows=48 width=262)
   ->  Seq Scan on dragons  (cost=0.00..1.11 rows=1 width=180)
         Filter: (sexe = 'M'::bpchar)
   ->  Bitmap Heap Scan on repas  (cost=4.16..9.50 rows=2 width=160)
         Recheck Cond: ((dragon)::text = (dragons.dragon)::text)
         ->  Bitmap Index Scan on keys  (cost=0.00..4.16 rows=2 width=0)
               Index Cond: ((dragon)::text = (dragons.dragon)::text)
(7 lignes)
```

Plan d'exécution d'une requête

Commande : explain

```
bpinaud=# explain analyze select * from repas natural join dragons where sexe='M';
                                                  QUERY PLAN
-------------------------------------------------------------------------------------------------------------------
Nested Loop  (cost=4.16..10.64 rows=48 width=262) (actual time=0.030..0.045 rows=9 loops=1)
->  Seq Scan on dragons  (cost=0.00..1.11 rows=1 width=180) (actual time=0.012..0.013 rows=6 loops=1)
      Filter: (sexe = 'M'::bpchar)
      Rows Removed by Filter: 3
->  Bitmap Heap Scan on repas  (cost=4.16..9.50 rows=2 width=160) (actual time=0.004..0.004 rows=2 loops=6)
      Recheck Cond: ((dragon)::text = (dragons.dragon)::text)
      Heap Blocks: exact=5
    -> Bitmap Index Scan on keys  (cost=0.00..4.16 rows=2 width=0) (actual time=0.002..0.002 rows=2 loops=6)
            Index Cond: ((dragon)::text = (dragons.dragon)::text)
Planning Time: 0.132 ms
Execution Time: 0.069 ms
```

Données Insee

Source des données : https://www.insee.fr/fr/statistiques/7632565

```
create table statistiques (
CODGEO varchar, REG int, DEP varchar, libgeo varchar,
P20_POP int, P14_POP int, P09_POP int, D99_POP int, D90_POP int, D82_POP int, D75_POP int,
D68_POP int,
SUPERF real,
NAIS1420 int, NAIS0914 int, NAIS9909 int, NAIS9099 int, NAIS8290 int, NAIS7582 int, NAIS6875 int,
DECE1420 int, DECE0914 int, DECE9909 int, DECE9099 int, DECE8290 int, DECE7582 int, DECE6875
int
);

\copy statistiques from 'base-cc-serie-historique-2020.csv' with (format 'csv', header, delimiter ';')
```

Plan d’exécution et index

bruno=> explain analyze select * from statistiques where p20_pop>100000;

```
                                                  QUERY PLAN
-------------------------------------------------------------------------------------------------------------
Seq Scan on statistiques  (cost=0.00..1108.60 rows=281 width=118) (actual time=0.434..5.019
rows=42 loops=1)
  Filter: (p20_pop > 100000)
  Rows Removed by Filter: 34886
Planning Time: 0.110 ms
Execution Time: 5.049 ms
(5 lignes)
```

Plan d'exécution et index

```
bruno=> explain analyze select * from statistiques where p20_pop>100000;
                                                 QUERY PLAN
-------------------------------------------------------------------------------------------------------------
Seq Scan on statistiques  (cost=0.00..1108.60 rows=281 width=118) (actual time=0.434..5.019
rows=42 loops=1)
  Filter: (p20_pop > 100000)
  Rows Removed by Filter: 34886
Planning Time: 0.110 ms
Execution Time: 5.049 ms
(5 lignes)
```

```
bruno=> create index pop20idx on statistiques (p20_pop);
CREATE INDEX
```

Plan d'exécution et index

```
...
Planning Time: 0.110 ms
Execution Time: 5.049 ms
(5 lignes)
bruno=> create index pop20idx on statistiques (p20_pop);
CREATE INDEX
bruno=> explain analyze select * from statistiques where p20_pop>100000;
                                                    QUERY PLAN
-------------------------------------------------------------------------------------------------------------------
Bitmap Heap Scan on statistiques  (cost=6.47..530.38 rows=281 width=118) (actual time=0.037..0.110 rows=42
loops=1)
  Recheck Cond: (p20_pop > 100000)
  Heap Blocks: exact=41
  ->  Bitmap Index Scan on pop20idx  (cost=0.00..6.40 rows=281 width=0) (actual time=0.015..0.016 rows=42
loops=1)
        Index Cond: (p20_pop > 100000)
Planning Time: 0.373 ms
Execution Time: 0.153 ms
(7 lignes)
```

Plan d'exécution et index

```
bruno=> explain analyze select * from statistiques where p20_pop between 1000 and 10000;
                                                                                                  QUERY PLAN
-------------------------------------------------------------------------------------------------------------------------------------------------
Bitmap Heap Scan on statistiques  (cost=139.80..945.72 rows=8928 width=118) (actual time=1.858..4.720 rows=8929 loops=1)
  Recheck Cond: ((p20_pop >= 1000) AND (p20_pop <= 10000))
  Heap Blocks: exact=654
  ->  Bitmap Index Scan on pop20idx  (cost=0.00..137.57 rows=8928 width=0) (actual time=1.573..1.573 rows=8929 loops=1)
        Index Cond: ((p20_pop >= 1000) AND (p20_pop <= 10000))
Planning Time: 0.199 ms
Execution Time: 5.358 ms
(7 lignes)
```

Plan d’exécution et index

bruno=> explain analyze select * from statistiques where p20_pop::real/d82_pop>1.5;

```
                                                        QUERY PLAN
---------------------------------------------------------------------------------------------------------------------------
Seq Scan on statistiques  (cost=0.00..1370.56 rows=11643 width=118) (actual time=0.005..2.344 rows=9288 loops=1)
  Filter: (((p20_pop)::real / (d82_pop)::double precision) > '1.5'::double precision)
  Rows Removed by Filter: 25636
Planning Time: 0.038 ms
Execution Time: 2.499 ms
(5 lignes)
```

Plan d’exécution et index

```
bruno=> explain analyze select * from statistiques where p20_pop::real/d82_pop>1.5;
                                                  QUERY PLAN
-------------------------------------------------------------------------------------------------------------------
Seq Scan on statistiques  (cost=0.00..1370.56 rows=11643 width=118) (actual time=0.005..2.344 rows=9288 loops=1)
  Filter: (((p20_pop)::real / (d82_pop)::double precision) > '1.5'::double precision)
  Rows Removed by Filter: 25636
Planning Time: 0.038 ms
Execution Time: 2.499 ms
(5 lignes)
```

```
bruno=> create index ratio_idx on statistiques ((p20_pop::real/d82_pop));
CREATE INDEX
```

Plan d'exécution et index

```
 ...
 Planning Time: 0.038 ms
 Execution Time: 2.499 ms
 (5 lignes)
bruno=> create index ratio_idx on statistiques ((p20_pop::real/d82_pop));
CREATE INDEX
bruno=> explain analyze select * from statistiques where p20_pop::real/d82_pop>1.5;
                                                        QUERY PLAN
-----------------------------------------------------------------------------------------------------------------------------
 Bitmap Heap Scan on statistiques  (cost=222.51..1127.33 rows=11641 width=118) (actual time=0.316..1.144
rows=9288 loops=1)
   Recheck Cond: (((p20_pop)::real / (d82_pop)::double precision) > '1.5'::double precision)
   Heap Blocks: exact=647
   ->  Bitmap Index Scan on ratio_idx  (cost=0.00..219.60 rows=11641 width=0) (actual time=0.264..0.264
rows=9288 loops=1)
         Index Cond: (((p20_pop)::real / (d82_pop)::double precision) > '1.5'::double precision)
 Planning Time: 0.044 ms
 Execution Time: 1.298 ms
(7 lignes)
```

Plan d’exécution et index

```
bruno=> explain analyze select * from statistiques where dep='31';
                                                        QUERY PLAN
---------------------------------------------------------------------------------------------------------------------------
 Seq Scan on statistiques  (cost=0.00..1108.55 rows=586 width=118) (actual time=0.764..2.330
rows=586 loops=1)
   Filter: ((dep)::text = '31'::text)
   Rows Removed by Filter: 34338
 Planning Time: 0.095 ms
 Execution Time: 2.368 ms
(5 lignes)
```

Plan d’exécution et index

```
bruno=> explain analyze select * from statistiques where dep='31';
                                                        QUERY PLAN
---------------------------------------------------------------------------------------------------------------------------
 Seq Scan on statistiques  (cost=0.00..1108.55 rows=586 width=118) (actual time=0.764..2.330
rows=586 loops=1)
   Filter: ((dep)::text = '31'::text)
   Rows Removed by Filter: 34338
 Planning Time: 0.095 ms
 Execution Time: 2.368 ms
(5 lignes)
bruno=> create index dep_idx on statistiques using hash(dep);
CREATE INDEX
bruno=>
```

Plan d'exécution et index

```
...
 Planning Time: 0.095 ms
 Execution Time: 2.368 ms
(5 lignes)
bruno=> create index dep_idx on statistiques using hash(dep);
CREATE INDEX
bruno=> explain analyze select * from statistiques where dep='31';
                                                 QUERY PLAN
-------------------------------------------------------------------------------------------------------------
 Bitmap Heap Scan on statistiques  (cost=20.54..706.62 rows=586 width=118) (actual time=0.021..0.068 rows=586
loops=1)
   Recheck Cond: ((dep)::text = '31'::text)
   Heap Blocks: exact=12
   ->  Bitmap Index Scan on dep_idx  (cost=0.00..20.39 rows=586 width=0) (actual time=0.014..0.015 rows=586
loops=1)
         Index Cond: ((dep)::text = '31'::text)
 Planning Time: 0.085 ms
 Execution Time: 0.087 ms
(7 lignes)
```

Plan d’exécution et index

```
bruno=> explain analyze select * from statistiques where libgeo like 'Saint%';
                                                      QUERY PLAN
-----------------------------------------------------------------------------------------------------------------------
Seq Scan on statistiques  (cost=0.00..1108.55 rows=3751 width=118) (actual time=0.019..2.941 rows=3911 loops=1)
  Filter: ((libgeo)::text ~~ 'Saint%'::text)
  Rows Removed by Filter: 31013
Planning Time: 0.090 ms
Execution Time: 3.038 ms
(5 lignes)
```

Plan d’exécution et index

```
bruno=> explain analyze select * from statistiques where libgeo like 'Saint%';
                                                    QUERY PLAN
-------------------------------------------------------------------------------------------------------------------
Seq Scan on statistiques  (cost=0.00..1108.55 rows=3751 width=118) (actual time=0.019..2.941
rows=3911 loops=1)
  Filter: ((libgeo)::text ~~ 'Saint%'::text)
  Rows Removed by Filter: 31013
Planning Time: 0.090 ms
Execution Time: 3.038 ms
(5 lignes)

bruno=> create index lib_dep_idx on statistiques (libgeo);
CREATE INDEX
bruno=>
```

Plan d’exécution et index

```
...
Planning Time: 0.090 ms
Execution Time: 3.038 ms
(5 lignes)
bruno=> create index lib_dep_idx on statistiques (libgeo);
CREATE INDEX
bruno=> explain analyze select * from statistiques where libgeo like 'Saint%';
                                                    QUERY PLAN
-------------------------------------------------------------------------------------------------------------------
Seq Scan on statistiques  (cost=0.00..1108.55 rows=3751 width=118) (actual time=0.039..3.341 rows=3911
loops=1)
  Filter: ((libgeo)::text ~~ 'Saint%'::text)
  Rows Removed by Filter: 31013
Planning Time: 0.178 ms
Execution Time: 3.431 ms
(5 lignes)
```

```
bruno=> create index lib_dep_idx2 on statistiques (libgeo varchar_pattern_ops);
CREATE INDEX
bruno=> explain analyze select * from statistiques where libgeo like 'Saint%';
                                                        QUERY PLAN
---------------------------------------------------------------------------------------------------------------------------
Bitmap Heap Scan on statistiques  (cost=98.06..816.10 rows=3751 width=118) (actual time=0.226..0.651 rows=3911
loops=1)
  Filter: ((libgeo)::text ~~ 'Saint%'::text)
  Heap Blocks: exact=194
  ->  Bitmap Index Scan on lib_dep_idx2  (cost=0.00..97.12 rows=3683 width=0) (actual time=0.207..0.208
rows=3911 loops=1)
        Index Cond: (((libgeo)::text ~>=~ 'Saint'::text) AND ((libgeo)::text ~<~ 'Sainu'::text))
Planning Time: 0.114 ms
Execution Time: 0.722 ms
(7 lignes)
```

```
bruno=> explain analyze select * from statistiques where libgeo like 'Saint%' and dep='31';
                                                
```

bruno=> drop index ….

bruno=> explain analyze select * from statistiques where libgeo like 'Saint%' and dep='31';

```
                                                    QUERY PLAN
-------------------------------------------------------------------------------------------------------------------
 Seq Scan on statistiques  (cost=0.00..1195.86 rows=63 width=118) (actual time=0.633..1.856 rows=53 loops=1)
   Filter: (((libgeo)::text ~~ 'Saint%'::text) AND ((dep)::text = '31'::text))
   Rows Removed by Filter: 34871
 Planning Time: 0.078 ms
 Execution Time: 1.866 ms
(5 lignes)
```

```
bruno=> explain analyze select * from statistiques where libgeo like 'Saint%' and dep='31';
                                                    QUERY PLAN
-------------------------------------------------------------------------------------------------------------------
 Seq Scan on statistiques  (cost=0.00..1195.86 rows=63 width=118) (actual time=0.633..1.856 rows=53 loops=1)
   Filter: (((libgeo)::text ~~ 'Saint%'::text) AND ((dep)::text = '31'::text))
   Rows Removed by Filter: 34871
 Planning Time: 0.078 ms
 Execution Time: 1.866 ms
(5 lignes)

bruno=> create index lib_dep_idx2 on statistiques (libgeo varchar_pattern_ops,dep);
CREATE INDEX
bruno=> explain analyze select * from statistiques where libgeo like 'Saint%' and dep='31';
                                                      QUERY PLAN
-----------------------------------------------------------------------------------------------------------------------------
 Bitmap Heap Scan on statistiques  (cost=114.34..301.49 rows=63 width=118) (actual time=0.288..0.294 rows=53 loops=1)
   Recheck Cond: ((dep)::text = '31'::text)
   Filter: ((libgeo)::text ~~ 'Saint%'::text)
   Heap Blocks: exact=2
   ->  Bitmap Index Scan on lib_dep_idx2  (cost=0.00..114.33 rows=62 width=0) (actual time=0.270..0.270 rows=53 loops=1)
         Index Cond: (((libgeo)::text ~>=~ 'Saint'::text) AND ((libgeo)::text ~<~ 'Sainu'::text) AND ((dep)::text = '31'::text))
 Planning Time: 0.168 ms
 Execution Time: 0.310 ms
(8 lignes)
```

bruno=> create index lib_dep_idx3 on statistiques (libgeo varchar_pattern_ops) where dep='31';
CREATE INDEX

```
bruno=> create index lib_dep_idx3 on statistiques (libgeo varchar_pattern_ops) where dep='31';
CREATE INDEX

bruno=> explain analyze select * from statistiques where libgeo like 'Saint%' and dep='31';
                                                        QUERY PLAN
---------------------------------------------------------------------------------------------------------------------------
Bitmap Heap Scan on statistiques  (cost=4.91..192.06 rows=63 width=118) (actual time=0.084..0.111 rows=53 loops=1)
  Recheck Cond: ((dep)::text = '31'::text)
  Filter: ((libgeo)::text ~~ 'Saint%'::text)
  Heap Blocks: exact=2
  ->  Bitmap Index Scan on lib_dep_idx3  (cost=0.00..4.90 rows=62 width=0) (actual time=0.063..0.064 rows=53 loops=1)
        Index Cond: (((libgeo)::text ~>=~ 'Saint'::text) AND ((libgeo)::text ~<~ 'Sainu'::text))
Planning Time: 1.853 ms
Execution Time: 0.173 ms
(8 lignes)
```

bruno=> explain analyze select dep,count(*) from statistiques where p20_pop < 1000 group by dep order by 2 desc;

```
                                                                 QUERY PLAN
---------------------------------------------------------------------------------------------------------------------------------------------
Sort  (cost=1237.59..1237.84 rows=100 width=11) (actual time=8.377..8.379 rows=95 loops=1)
  Sort Key: (count(*)) DESC
  Sort Method: quicksort  Memory: 28kB
  ->  HashAggregate  (cost=1233.27..1234.27 rows=100 width=11) (actual time=8.356..8.362 rows=95 loops=1)
        Group Key: dep
        Batches: 1  Memory Usage: 24kB
        ->  Seq Scan on statistiques  (cost=0.00..1108.55 rows=24944 width=3) (actual time=0.018..3.947
rows=24982 loops=1)
              Filter: (p20_pop < 1000)
              Rows Removed by Filter: 9942
Planning Time: 0.205 ms
Execution Time: 8.433 ms
(11 lignes)
```

bruno=> explain analyze select dep,count(*) from statistiques where p20_pop < 1000 group by dep order by 2 desc limit 3;

```
                                                
```

```
bruno=> explain analyze select dep,count(*) from statistiques where p20_pop > 1000 group by dep order by 2 desc limit 3;
                                                                                  QUERY PLAN
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------
Limit  (cost=982.45..982.46 rows=3 width=11) (actual time=6.406..6.407 rows=3 loops=1)
  ->  Sort  (cost=982.45..982.70 rows=100 width=11) (actual time=6.404..6.405 rows=3 loops=1)
        Sort Key: (count(*)) DESC
        Sort Method: top-N heapsort  Memory: 25kB
        ->  HashAggregate  (cost=980.16..981.16 rows=100 width=11) (actual time=6.385..6.392 rows=100 loops=1)
              Group Key: dep
              Batches: 1  Memory Usage: 24kB
              ->  Bitmap Heap Scan on statistiques  (cost=133.60..930.28 rows=9975 width=3) (actual time=1.976..3.925 rows=9935
loops=1)
                    Recheck Cond: (p20_pop > 1000)
                    Heap Blocks: exact=654
                    ->  Bitmap Index Scan on p20idx  (cost=0.00..131.10 rows=9975 width=0) (actual time=1.698..1.698 rows=9935 loops=1)
                          Index Cond: (p20_pop > 1000)
Planning Time: 0.223 ms
Execution Time: 6.467 ms
(14 lignes)
```

```
bruno=> create table foo(a int primary key, b int unique, c int);
CREATE TABLE
bruno=> \d foo
                Table « public.foo »
Colonne |  Type   | Collationnement | NULL-able | Par défaut
---------+---------+-----------------+-----------+------------
a       | integer |                 | not null  |
b       | integer |                 |           |
c       | integer |                 |           |
Index :
   "foo_pkey" PRIMARY KEY, btree (a)
   "foo_b_key" UNIQUE CONSTRAINT, btree (b)
```